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PROJECT REPORT ENTITLED

DYNAMIC CONTEXT-AWARE TRIES FOR KNOWLEDGE GRAPH-CONSTRAINED LARGE LANGUAGE MODEL GENERATION

BY

BERNARD KIRK ADJANOR KATAMANSO — FOE.41.008.088.22

AMONOR ERICA — FOE.41.008.030.22

JOSEPH OSEI NYARKO — FOE.41.008.113.22

JESSICA AFUA ETORNAM NSAFOAH — FOE.41.008.107.22

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PROJECT SUPERVISOR

.....

Dr ERIC AFFUM

Tarkwa, Ghana

Month, Year

DECLARATION

We, BERNARD KIRK ADJANOR KATAMANSO, AMONOR ERICA, JOSEPH OSEI NYARKO and JESSICA AFUA ETORNAM NSAFOAH, declare that this project work is our own original work. It is being submitted for the degree of Bachelor of Science in Computer Science and Engineering at the University of Mines and Technology (UMaT), Tarkwa. It has not been submitted, either in whole or in part, for any degree or examination at any other university.

Signature of Student: _____

BERNARD KIRK ADJANOR KATAMANSO (FOE.41.008.088.22)

Signature of Student: _____

AMONOR ERICA (FOE.41.008.030.22)

Signature of Student: _____

JOSEPH OSEI NYARKO (FOE.41.008.113.22)

Signature of Student: _____

JESSICA AFUA ETORNAM NSAFOAH (FOE.41.008.107.22)

_____ day of _____ (year) _____

ABSTRACT

Large language models (LLMs) are capable of multi-hop reasoning but remain prone to hallucination. Knowledge graph-constrained decoding mitigates this by restricting output tokens to valid paths within a knowledge graph (KG), ensuring absolute structural faithfulness, unlike standard soft-grounding approaches such as Retrieval-Augmented Generation. However, leading constrained paradigms, including Graph-Constrained Reasoning (GCR), construct constraints solely from graph structure. By ignoring the question's semantics and the evolving generation state, these approaches suffer from the *permissiveness problem*: at every decoding step, the allowed valid token set contains thousands of structurally accessible but contextually irrelevant paths. This excess of valid tokens forces the LLM to rely on the very parametric biases that constrained decoding was designed to suppress.

To resolve this limitation, we propose Dynamic Context-Aware Tries (DCA-Trie), a novel framework that conditions the valid tokens jointly on the knowledge graph, the input question, and the partially generated reasoning chain. This approach upgrades the constraint oracle to an active semantic state $f(\mathcal{G}, E_q, q, y_{<t})$ by using a pretrained relevance scorer to filter candidate KG paths. We develop two distinct architectures: static semantic filtering applied at trie construction (DCA-Trie v1), and dynamic, step-wise bounding conditioned on the evolving generation (DCA-Trie v2).

Both variants are evaluated on the WebQSP and ComplexWebQuestions datasets over Freebase. Alongside standard evaluating metrics (Hits@1, F1), we introduce the Semantic Irrelevance Ratio (SIR), the first metric designed to uniquely isolate and quantify the proportion of structurally valid but semantically irrelevant paths in a constraint set.

DEDICATION

To our parents, who taught us that the truest things cannot be computed.

And to the machine, an elegant engine of probability, reminding us that predicting the word is not understanding the world.

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LIST OF ABBREVIATIONS

DCA-Trie	Dynamic Context-Aware Trie
LLM	Large Language Model
KGQA	Knowledge Graph Question Answering
RAG	Retrieval-Augmented Generation
KG	Knowledge Graph
GCR	Graph-Constrained Reasoning
CWQ	ComplexWebQuestions
RLHF	Reinforcement Learning from Human Feedback
CoT	Chain-of-Thought
DoG	Decoding on Graphs
SIR	Semantic Irrelevance Ratio
GCD	Grammar-Constrained Decoding
QA	Question Answering
NLP	Natural Language Processing
AI	Artificial Intelligence
BFS	Breadth-First Search

CHAPTER 1

INTRODUCTION

1.1 Background

Large language models (LLMs) have emerged as one of the most consequential advances in artificial intelligence, demonstrating capabilities across language understanding, generation, and multistep reasoning that were considered beyond reach only a decade ago (Brown *et al.*, 2020; OpenAI, 2024). Models such as the GPT series (Brown *et al.*, 2020; OpenAI, 2024) and the Llama family (Touvron *et al.*, 2023; Grattafiori *et al.*, 2024) achieve strong performance on a broad spectrum of language tasks by learning statistical regularities over vast corpora of text. The scale and breadth of this training enables these models to exhibit apparent knowledge across diverse domains, from history and science to legal reasoning and medical information, without any explicit representation of factual knowledge in their architecture.

Despite these achievements, a fundamental and persistent failure mode undermines the reliability of LLMs in knowledge-intensive settings: hallucination. Ji *et al.* (2023) define hallucination as the generation of content that is fluent and coherent but factually unsupported or contradicted by verifiable sources. Huang *et al.* (2023b) provide a comprehensive taxonomy distinguishing intrinsic hallucination, where the output contradicts a provided source, from extrinsic hallucination, where the output cannot be verified against any source. Critically, hallucination is not a rare edge case. Luo *et al.* (2024b) report that state-of-the-art retrieval-augmented approaches produce hallucinated reasoning steps in approximately one-third of cases even when granted access to a supporting knowledge graph. This failure rate renders unconstrained LLMs unsuitable for high-stakes knowledge-intensive tasks without additional mechanisms to enforce factual grounding. The structural inevitability of hallucination has been formally established in recent theoretical work. Xu *et al.* (2025) prove that for any computable ground truth function, no LLM training procedure can achieve zero hallucination probability across all inputs–hallucination is an inherent property of computable learning systems rather than a transient training artifact. This "inevitability" result underscores that mitigation strategies relying solely on improved training, scaling, or prompting cannot fully eliminate hallucination; external structural constraints that ground generation in verifiable

sources are theoretically necessary for guaranteed faithfulness. Knowledge graph-constrained decoding provides exactly such an external mechanism, making hallucination mathematically impossible rather than merely improbable.

The structural cause of hallucination is well understood. Autoregressive LLMs generate text by conditioning each token on all previously generated tokens, producing a probability distribution over the vocabulary at each step: $P(y_t \mid y_{<t}, x)$, where x is the input and $y_{<t}$ is the partially generated sequence (Brown *et al.*, 2020). This process is effective at producing locally coherent text but provides no mechanism for grounding each generated claim in a verifiable external source. An error introduced at step t propagates forward as the conditioning context for all subsequent tokens, producing confident and grammatically well-formed outputs whose factual incorrectness has no surface signal (Huang *et al.*, 2023b; Ji *et al.*, 2023).

Knowledge graph question answering (KGQA) represents one of the most demanding contexts in which this failure mode manifests. A knowledge graph (KG) is a structured, human-curated database of verified facts encoded as (entity, relation, entity) triplets, for instance, (Marie Curie, award received, Nobel Prize in Physics) or (Paris, capital of, France), forming a large-scale graph over which multi-hop reasoning paths can be traced (Bollacker *et al.*, 2008). The Freebase knowledge graph, which provides the underlying resource for the standard KGQA benchmarks WebQSP (Yih *et al.*, 2016a) and ComplexWebQuestions (Talmor and Berant, 2018a), encodes tens of millions of such triplets covering a wide range of domains and relation types. Answering a multi-hop question such as 'What is the nationality of the director of the film in which the Blue Hawaii actor starred?' requires traversing a chain of three or more KG triplets in the correct order, a task for which an unconstrained LLM, relying entirely on parametric memory, may produce a plausible-sounding but factually incorrect chain that is indistinguishable in fluency from a correct one (Lan *et al.*, 2022a; He *et al.*, 2021a).

The dominant paradigm for addressing this problem is retrieval-augmented generation (RAG), in which relevant passages or subgraphs are retrieved from an external source and provided to the LLM as additional context (Lewis *et al.*, 2020b; Izacard and Grave, 2021). While RAG reduces hallucination by providing relevant source material, it does not prevent the LLM from ignoring that material or generating claims that go beyond it. The LLM remains free to

produce any token at any step; the retrieved context is a soft influence on its output, not a hard constraint on what it may generate (Asai *et al.*, 2024; Agrawal *et al.*, 2024b).

KG-constrained decoding addresses this limitation directly. Rather than providing the KG as a soft contextual influence, constrained decoding frameworks interpose a constraint oracle between the LLM's logit distribution and the token selection mechanism. At each generation step, the oracle computes a binary mask over the vocabulary, setting to zero the logit of any token that would not correspond to a valid continuation of a KG path. The masking formulation, introduced by Geng *et al.* (2023) in the context of grammar-constrained decoding, takes the form $\tilde{\alpha}_t = m_t \odot \alpha_t$, where α_t is the raw logit vector, $m_t \in \{0, 1\}^{|V|}$ is the validity mask, and $\odot$ denotes element-wise product. The generation process thus selects from a restricted set of valid tokens at every step, making structural hallucination, the generation of facts that do not exist in the KG, mathematically impossible rather than merely improbable (Willard and Louf, 2023; Geng *et al.*, 2023).

The leading implementation of this paradigm is Graph-Constrained Reasoning (GCR) (Luo *et al.*, 2024b), which achieves 92.6 Hits@1 on WebQSP and 75.8 Hits@1 on CWQ with verified 100% structural faithfulness. GCR constructs a KG-Trie, a prefix tree (Fredkin, 1960) over all KG paths within L hops of the question entities, obtained by breadth-first search, and uses it as the constraint oracle during beam-search decoding. Because every token in the generated reasoning chain must correspond to a valid prefix in the trie, the output is guaranteed to consist entirely of verified KG facts. This represents a qualitative improvement over both unconstrained LLMs and soft RAG approaches for the specific property of structural faithfulness.

However, a significant limitation remains. The KG-Trie in GCR is computed once, before generation begins, from the graph structure and the question entity set alone. It carries no information about the question semantics, the direction in which reasoning is proceeding, or which entities and relations have already been committed to in the partial generation $y_{<t}$. Formally, the valid token set W_{val} is a function $f(G, E_q)$, independent of q and $y_{<t}$, and remains identical across every generation step. For a question about a specific narrow fact, this trie may contain hundreds or thousands of structurally valid but semantically irrelevant paths, forcing the LLM to discriminate among them using its own parametric knowledge: precisely the hallucination-prone mechanism that constrained decoding was designed to

suppress (Li *et al.*, 2024; Luo *et al.*, 2024b).

Decoding on Graphs (DoG) (Li *et al.*, 2024) introduces dynamic step-wise expansion, updating the constraint trie after each committed entity to include KG neighbors of the current node. This topological dynamism ensures the constraint set reflects the reasoning direction taken, addressing GCR's structural rigidity. DoG also employs question semantics, but crucially, only for *ranking* candidate paths during beam search–the valid token set itself remains determined solely by graph connectivity, admitting every neighbor regardless of relevance to the question or partial generation. The permissiveness problem therefore persists: thousands of structurally valid but semantically irrelevant paths remain in the valid set at each step, forcing the LLM to discriminate among them using parametric knowledge. Furthermore, DoG sacrifices the computational efficiency of pre-computed tries, rebuilding the constraint structure at each decoding step. These limitations motivate the DCA-Trie framework, which applies semantic filtering directly to the constraint oracle, ensuring that the valid token set at each step is both structurally faithful and semantically relevant.

1.2 Problem Statement

Existing KG-constrained decoding frameworks, including GCR (Luo *et al.*, 2024b) and DoG (Li *et al.*, 2024), construct their constraint oracles as functions $f(G, E_q)$ of the knowledge graph and question entities, evaluated prior to or independently of the sequential generation process. This design has been shown to guarantee structural faithfulness (every generated token corresponds to a verified KG triplet), but provides no mechanism for adapting the constraint to the semantic content of the question or the direction taken by the partial generation (Geng *et al.*, 2023; Willard and Louf, 2023).

The consequence is a fundamental conflation of two distinct requirements. Structural validity requires that every generated token correspond to a valid KG path prefix, a requirement all existing frameworks satisfy. Contextual relevance requires that the constraint set reflect the semantic focus of the reasoning at the current step, a requirement no existing framework satisfies. As a result, existing constraint oracles are simultaneously too broad, admitting structurally valid but semantically irrelevant paths that force the model to rely on parametric discrimination, and too rigid, failing to narrow in response to the evolving reasoning direction as generation progresses (Li *et al.*, 2024; Luo *et al.*, 2024b).

This project addresses the following specific problem: current KG-constrained decoding frameworks do not condition the valid token constraint on the input question semantics or the partial generation, leaving the LLM responsible for semantic discrimination among a large set of valid-but-irrelevant paths at every generation step. This residual reliance on parametric knowledge partially re-introduces the hallucination risk that constrained decoding was designed to eliminate, and is most pronounced in multi-hop settings where the number of valid paths grows exponentially with hop depth (Lan *et al.*, 2022a; He *et al.*, 2021a; Talmor and Berant, 2018a).

Very recent work has independently addressed the efficiency-faithfulness trade-off in KGQA. RouterKGQA (Yuan *et al.*, 2026) proposes routing between specialized and general models with semantic filtering at the answer level, achieving strong performance with minimal inference cost. While RouterKGQA and DCA-Trie both employ semantic relevance scoring, they operate at fundamentally different architectural stages: RouterKGQA filters *answers* after generation, whereas DCA-Trie filters *tokens* during generation. This distinction is critical: DCA-Trie's constraint oracle ensures the LLM never sees semantically irrelevant tokens, reducing cognitive load and enabling deterministic faithfulness guarantees that post-hoc filtering cannot provide. Furthermore, DCA-Trie introduces the Semantic Irrelevance Ratio (SIR; Chapter **??**), the first metric to quantify constraint permissiveness independently of downstream accuracy–an evaluation dimension not addressed by routing-based approaches.

To address these limitations, this project proposes DCA-Trie, upgrading the constraint oracle from $W_{\text{val}}(t) = f(G, E_q)$ to $W_{\text{val}}(t) = f(G, E_q, q, y_{<t})$. The upgraded framework must simultaneously satisfy three conditions: (i) structural faithfulness is preserved, every token admitted at each step corresponds to a valid KG path, as verified by exhaustive triplet checking against Freebase; (ii) semantic irrelevance is reduced, the proportion of the valid set consisting of paths irrelevant to the question decreases relative to GCR and DoG, as measured by the Semantic Irrelevance Ratio (SIR) introduced in this work; and (iii) answer recall is maintained: no gold answer path is erroneously excluded by the relevance filter, with false negative rate held below 5% on the WebQSP validation set.

1.3 Aim and Objectives

This project aims to design, implement, and evaluate a Dynamic Context-Aware Trie (DCA-Trie) framework for knowledge graph-constrained large language model generation, in which the valid token constraint at each decoding step is conditioned jointly on the knowledge graph structure, the input question semantics, and the partially generated reasoning chain, achieving improved answer accuracy on multi-hop KGQA benchmarks while maintaining 100% structural faithfulness to the underlying knowledge graph.

To achieve this aim, the following objectives have been set:

i. To characterize the permissiveness problem in existing KG-constrained decoding frameworks by defining and measuring the Semantic Irrelevance Ratio (SIR) of GCR's static KG-Tries on the WebQSP and CWQ benchmarks, stratified by hop depth.

ii. To design a semantic relevance scoring mechanism using pretrained sentence transformer embeddings (Reimers and Gurevych, 2019a) that can filter candidate KG paths by joint relevance to the input question and partial generation at inference time, without task-specific fine-tuning.

iii. To implement a static semantic filtering variant (DCA-Trie v1) that applies the relevance scorer at KG-Trie construction time, reducing trie size before generation begins while maintaining a false negative rate below 5% on gold answer paths on the WebQSP validation set.

iv. To implement a step-wise dynamic expansion variant (DCA-Trie v2) in which the trie is updated after each committed triplet using the relevance scorer conditioned jointly on the question and the partial generation $y_{<t}$, directly operationalizing the $W_{\text{val}}(t) = f(G, E_q, q, y_{<t})$ formulation.

v. To evaluate DCA-Trie v1 and DCA-Trie v2 against the GCR baseline and an unconstrained chain-of-thought (CoT) baseline on WebQSP and CWQ, measuring Hits@1, F1, structural faithfulness, SIR, and average trie size, with results stratified by question hop depth.

vi. To demonstrate the practical applicability of DCA-Trie by building an experimental interactive prototype system (web-based or command-line QA interface) for research

demonstration, allowing users to input complex natural language questions and receive factually grounded reasoning chains and answers, thereby showcasing the framework's real-world value beyond benchmark metrics.

1.4 Tools and Facilities Used

1.4.1 Software Tools

Table 1.1 presents an evaluation of the software tools implemented in this project, their role within the DCA-Trie framework, and the rationale for their selection.

Table 1.1 Implemented Software Tools Evaluation

Tool	Use in Project	Rationale
Python / HuggingFace Transformers	Model inference pipeline; loading and running GCR's Llama-3.1-8B checkpoint; integrating the relevance scorer	Provides reliable model loading, tokenization, and generation utilities, and enables direct use of the released GCR checkpoint.
GCR Framework	Primary baseline; KG-Trie construction, graph-constrained decoding, and result reproduction on WebQSP and CWQ	Provides a strong baseline and supports the trie-based constrained decoding mechanism that DCA-Trie extends.
Sentence-transformers / all-MiniLM-L6-v2	Semantic relevance scoring; computing cosine similarity between question-context and candidate KG path embeddings	Provides fast semantic embeddings without extra fine-tuning and supports efficient iterative decoding.
PyTorch	Deep learning framework underpinning all model inference and custom decoding logic	Supports implementation and testing of custom decoding logic and integrates smoothly with HuggingFace.
Freebase / WebQSP and CWQ Datasets	Knowledge graph source and KGQA benchmarks for evaluation	Provides standard datasets for this research area and enables direct comparison with prior work.
Google Colab Pro (A100, 40 GB)	GPU environment for all inference experiments; Llama-3.1-8B constrained decoding with beam search	Provides enough GPU memory for stable experiments and removes the need for local high-end hardware.
Visual Studio Code + Git / GitHub	Development environment and version control for all four team members	Supports day-to-day coding, team collaboration, version tracking, and reproducibility throughout the project.

1.4.2 Facilities

The following facilities were utilized in the execution of this project:

i. Google Colab Pro (A100, 40 GB VRAM) for all model inference and constrained decoding experiments.

ii. University library and online academic databases (ACM Digital Library, arXiv, Google Scholar, ACL Anthology) for literature review and citation verification.

iii. University internet infrastructure for cloud connectivity, dataset access, and retrieval of model checkpoints from the HuggingFace model hub.

iv. Shared team repository hosted on GitHub for version control, experiment logging, and reproducibility of all reported results.

1.5 Methods Used

The project employs the following methods:

i. Systematic review and critical analysis of the academic literature on constrained decoding, knowledge graph question answering, hallucination in LLMs, and retrieval-augmented generation.

ii. Formal gap analysis of the four foundational papers, GCR (Luo *et al.*, 2024b), DoG (Li *et al.*, 2024), GCD (Geng *et al.*, 2023), and Outlines (Willard and Louf, 2023), to identify the specific limitations that motivate the DCA-Trie design, including the permissiveness problem and the absence of generation-state feedback in the constraint oracle.

iii. Formal specification of the DCA-Trie framework, including definition of the Semantic Irrelevance Ratio metric, the relevance scoring mechanism using sentence transformer embeddings (Reimers and Gurevych, 2019a), the static filtering procedure for v1, and the step-wise dynamic expansion rule for v2.

iv. Reproduction of the GCR baseline on the WebQSP benchmark using GCR's publicly released Llama-3.1-8B checkpoint to establish a verified starting point within 1–2% of published Hits@1.

v. Implementation of DCA-Trie v1 and v2 as extensions to GCR's inference pipeline, with relevance threshold selected on a 100-question held-out validation subset of WebQSP to minimize false negative rate on gold answer paths.

vi. Quantitative evaluation of all system configurations on WebQSP and CWQ using Hits@1, F1, faithfulness rate, SIR, and average trie size per step, stratified by question hop depth following the evaluation protocol of Lan *et al.* (2022a).

vii. Ablation studies examining the sensitivity of DCA-Trie's performance to the relevance threshold τ, the sentence encoder architecture, and beam size k, isolating the contribution of each mechanism to the overall result.

viii. Qualitative error analysis identifying failure cases in which DCA-Trie's semantic filter erroneously excludes a gold answer path, supporting honest characterization of the framework's limitations and providing concrete directions for future work.

1.6 Scope of Work

This project designs and evaluates a Dynamic Context-Aware Trie framework for KG-constrained LLM decoding, with the specific aim of addressing the permissiveness problem identified in GCR and DoG. The framework is developed and evaluated in the context of multi-hop KGQA over Freebase-grounded benchmarks (WebQSP and CWQ), which are the standard evaluation settings in the KG-constrained decoding literature. All experiments use GCR's publicly released Llama-3.1-8B checkpoint; no new model training or fine-tuning is performed. The semantic relevance scorer uses the all-MiniLM-L6-v2 sentence transformer without domain-specific adaptation. We emphasize that all experiments use GCR's publicly released Llama-3.1-8B checkpoint (Luo *et al.*, 2025), which includes their fine-tuned KG-specialized parameters. No additional model training or fine-tuning is performed in this work; our contribution is strictly the constraint oracle architecture and semantic filtering mechanism.

The scope is explicitly bounded as follows. Generalization of DCA-Trie to knowledge graphs other than Freebase, or to non-KGQA tasks such as entity linking or structured information extraction, is identified as future work and is not evaluated in this dissertation. Formal proofs of the faithfulness guarantee under dynamic pruning are beyond scope; empirical

measurement of the gold answer false negative rate is provided as evidence. The confidence-conditioned constraint tightening mechanism, in which constraint granularity is modulated by the LLM's token-level entropy, is described as a theoretical extension but is not implemented. Production-scale deployment, latency optimization for online serving, and integration with proprietary enterprise knowledge graphs are engineering extensions beyond the research scope of this project.

The primary contribution of this work is the DCA-Trie framework and the Semantic Irrelevance Ratio metric. Secondary contributions are the empirical characterization of constraint permissiveness in existing frameworks and the ablation analysis isolating the contribution of semantic filtering from dynamic expansion. All code, experimental configurations, and the curated evaluation datasets will be made publicly available to support reproducibility and future extension.

1.7 Organization of Work

Chapter 1 introduces the study by identifying the core problem, stating the project objectives, and defining the intended DCA-Trie solution, thereby establishing the motivation, scope, and overall direction of the research. Chapter 2 presents the literature review by examining prior work on hallucination in LLMs, KGQA, constrained decoding, and related frameworks, and it identifies key limitations in existing approaches to justify the need for a context-aware constraint mechanism. Chapter 3 combines problem formulation and methodology, defining the upgraded constraint setup, research questions, evaluation criteria, and implementation strategy for DCA-Trie v1 and v2 within the GCR pipeline (Luo *et al.*, 2024b). Chapter 4 presents results and discussion, reporting quantitative evaluation across WebQSP and CWQ, analysing faithfulness and semantic irrelevance trends, and interpreting the findings through ablation and error analysis. Chapter 5 concludes the dissertation by summarizing the main contributions, highlighting practical implications, and presenting recommendations for future work.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Large language models (LLMs) have fundamentally transformed natural language processing, demonstrating remarkable capabilities across a wide range of tasks. However, their reliance on autoregressive generation introduces a critical vulnerability: the propensity to confidently generate fluent but factually incorrect information, known as hallucination (Ji *et al.*, 2023). In domains requiring precise, verifiable reasoning, such as knowledge-intensive question answering, this lack of factual reliability remains a significant barrier. To address this, the field has increasingly looked toward grounding language models in structured, external facts using knowledge graphs. Yet, ensuring that a model remains strictly faithful to these graphs during generation, without sacrificing fluency, remains an open and complex challenge.

The purpose of this chapter is to survey the existing literature that contextualizes and motivates the research problem this dissertation addresses: how to structurally guarantee factual faithfulness in multi-hop language model reasoning. Rather than describing the proposed system itself, this review maps the intersecting research landscapes of language model hallucination, multi-step reasoning, and structured knowledge grounding. By critically examining current mitigation strategies, ranging from advanced prompting to retrieval-augmented generation and early constrained decoding frameworks, this chapter establishes the fundamental limitations of approaches that rely solely on soft grounding or static constraints. Ultimately, it defines the specific boundaries where this dissertation makes its contribution.

The review is structured as follows. Section 2.2 covers the fundamentals of Large Language Models. Section 2.3 establishes hallucination in large language models as a structural problem inherent to autoregressive generation. Section 2.4 examines chain-of-thought reasoning and its extensions, showing why improved prompting strategies cannot provide structural faithfulness guarantees. Section 2.5 introduces knowledge graphs and knowledge graph question answering, including the benchmarks and the complexity properties that make multi-hop KGQA challenging. Section 2.6 traces the development of constrained decoding from foundational logit masking through grammar constraints to knowledge graph-specific

frameworks. Section 2.7 reviews retrieval-augmented generation and establishes why soft grounding approaches cannot achieve structural faithfulness. Section 2.8 reviews semantic similarity scoring with sentence transformers. Section 2.9 surveys related works organized by approach and critiques each cluster. Section 2.10 synthesizes the three gaps that remain unresolved across the literature.

2.2 Large Language Models

To understand how knowledge graph constraints can be applied to language models, it is first necessary to understand the underlying mechanism of language model generation. Large language models (LLMs) are the foundational engines driving modern natural language processing. This section establishes the architectural and generative properties of LLMs, specifically detailing the autoregressive process that gives rise to both their fluent reasoning capabilities and their vulnerability to factual hallucination.

2.2.1 Transformer Architecture

Modern LLMs are built upon the Transformer architecture, introduced by Vaswani *et al.* (2017). At its core, the Transformer relies on a self-attention mechanism that allows the model to dynamically weigh the importance of different tokens in an input sequence regardless of their positional distance. This represented a fundamental shift from prior recurrent architectures, enabling highly parallelized training and the capture of long-range dependencies. While the original framework proposed an encoder-decoder structure designed for translation tasks, the field has largely converged on decoder-only variants for text generation. In a decoder-only model, self-attention is causally masked, meaning that when computing the representation for a given token, the model can only attend to previous tokens and not future ones. This strictly left-to-right processing naturally aligns with the task of predicting the next word in a sequence. The Llama-3.1-8B model (Dubey *et al.*, 2024), which serves as the base generator in this project along with earlier frameworks, is precisely this kind of decoder-only, causally-masked transformer.

2.2.2 Autoregressive Generation

The text production process in these models is fundamentally autoregressive. Given a vocabulary $\mathcal{V}$, an input sequence x, and previously generated output tokens $y_{<t}$, the model

computes a probability distribution over the entire vocabulary for the next token: $P(y_t|y_{<t}, x)$. Because the generation condition includes all previously generated tokens, every new word becomes part of the context for the next word. Decoding strategies, such as beam search and nucleus sampling (Holtzman *et al.*, 2020), determine how a specific token is selected from this computed distribution. Crucially, because this distribution is always computed over the full vocabulary, any mathematically possible token has a strictly positive, non-zero probability of being generated at any step. This autoregressive property is the mechanism that produces highly fluent, coherent text; however, because each step conditions on prior outputs without an external verifier, it is also the direct structural cause of hallucination.

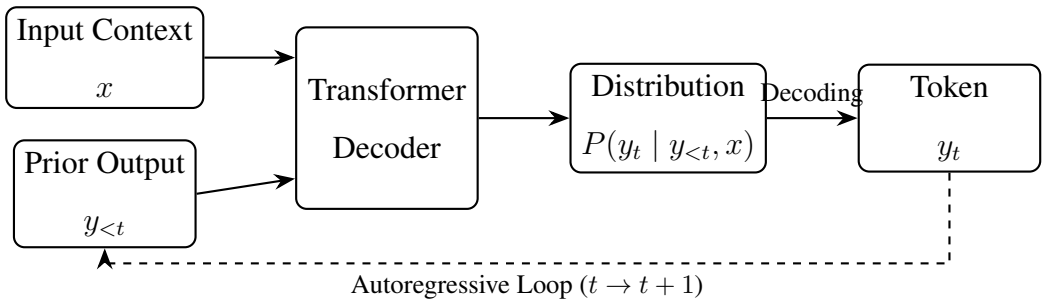

Figure 2.1 The autoregressive generation process.

2.2.3 Large-Scale Pretraining

The transition from basic transformers to "large" language models is driven by scale in both parameter count and training data. Foundational models are pretrained on massive corpora of unlabelled text using a simple self-supervised objective, typically predicting the next token. The GPT series (Brown *et al.* 2020; Achiam *et al.* 2023) and the open-weight Llama series (Touvron *et al.* 2023; Dubey *et al.* 2024) demonstrated that scaling this simple objective leads to the emergence of powerful downstream capabilities, including few-shot learning and broad world knowledge. During pretraining, the model internalises patterns, facts, and logical structures present in its training distribution. However, this parametric memory is fundamentally static and statistical. While scaling up models improves their fluency, syntax, and ability to recall commonly observed facts, it does not fix the underlying generation mechanism. The model still lacks a structural way to interact with dynamic external facts or guarantee that its statistical outputs map to verified truth in the real world.

2.2.4 Instruction Following and Chain-of-Thought

To align these pretrained base models with user intent, they undergo post-training processes such as instruction tuning and reinforcement learning from human feedback (RLHF). This alignment allows models to engage in dialogue and follow complex instructions. One of the most significant capabilities elicited from aligned LLMs is multi-step reasoning, typically prompted via Chain-of-Thought (CoT; Wei *et al.* 2022). As surveyed by Huang and Chang (2023), CoT prompting encourages the model to generate intermediate reasoning steps before arriving at a final answer, effectively breaking down complex problems. While CoT dramatically improves performance on logic and reasoning benchmarks, it suffers from the same autoregressive limitation: each intermediate reasoning step is generated using only the model's internal statistical distributions. Therefore, while CoT helps structure a language model's reasoning process, it cannot logically ground those outputs in verifiable sources. This persistent lack of factual grounding creates the necessity for more rigorous interventions, such as knowledge graph-constrained decoding.

2.3 Hallucination in Large Language Models

2.3.1 Definition and Taxonomy

Hallucination in natural language generation refers to the production of content that is fluent and internally coherent yet factually unsupported by or in outright contradiction with verifiable sources (Ji *et al.*, 2023). Huang *et al.* (2023a) refine this definition into a two-part taxonomy. *Intrinsic hallucination* occurs when the model output directly contradicts material present in the provided source context. *Extrinsic hallucination* occurs when the output contains claims that cannot be verified against any external source, even if not directly contradicting a specific reference.

Both types are documented at non-trivial rates in deployed systems. Ji *et al.* (2023) survey over 30 works across summarization, dialogue, and question answering, finding hallucination rates ranging from 15% to over 60% depending on task type and measurement methodology. In the specific context of knowledge graph question answering, Luo *et al.* (2024b) report that retrieval-augmented reasoning approaches produce hallucinated reasoning steps in approximately one-third of cases even when provided direct access to a supporting knowledge graph, underscoring that the problem is not simply one of insufficient external knowledge.

2.3.2 Structural Causes: The Autoregressive Generation Mechanism

The root cause of hallucination lies in the autoregressive generation mechanism shared by all modern large language models. At each generation step, the model computes a probability distribution over its vocabulary conditioned on the input and all previously generated tokens. Three structural properties of this mechanism make hallucination an ever-present risk rather than an occasional failure.

First, the generation distribution is computed entirely from learned parameters, with no term that corresponds to a factual verification operation against an external source. The model has no mechanism to "check" a candidate token against a knowledge base before selecting it. Second, errors propagate forward: because each step is conditioned on all prior output, a hallucinated token at step t becomes part of the conditioning context for step $t+1$, producing fluent and internally consistent continuations of an already-incorrect chain. Third, the distribution is defined over the full vocabulary at every step, meaning that any token has a strictly positive probability of selection at any step, regardless of factual correctness (Brown *et al.*, 2020; Huang *et al.*, 2023a; Ji *et al.*, 2023).

2.3.3 The Scaling Paradox

A common objection to the severity of the hallucination problem is that sufficiently large models should eventually achieve adequate factual reliability. The evidence does not support this view. Lin *et al.* (2022) introduce TruthfulQA, a benchmark targeting questions that frequently elicit plausible but false responses. Their central finding is that larger models do not outperform smaller models on truthfulness and in certain categories exhibit *inverse scaling*; in other words, larger models perform worse because they more effectively reproduce prevalent but incorrect beliefs from training data.

Perez *et al.* (2023) identify a compounding problem in models trained with reinforcement learning from human feedback: such models learn to generate confident-sounding responses that align with apparent user expectations rather than with verifiable facts, decoupling confidence from accuracy. Kandpal *et al.* (2023) show that while larger models memorize more training-data facts and improve on frequently-seen entity combinations, they do not improve on compositional questions requiring combinations of facts not seen during training. This precisely mirrors the structure of multi-hop knowledge graph questions. Mallen *et al.*

(2023) confirm substantial hallucination rates for GPT-4 on questions involving tail entities, regardless of model scale. Together, these findings establish that hallucination is a structural feature of the autoregressive mechanism, not a scale problem with a straightforward remedy.

2.4 Multi-Step Reasoning: Chain-of-Thought and Its Limits

2.4.1 Chain-of-Thought Prompting

Chain-of-thought (CoT) prompting, introduced by Wei *et al.* (2022), is the dominant paradigm for eliciting multi-step reasoning from large language models. By including worked examples in the prompt that show intermediate reasoning steps, the model learns to decompose complex problems into a sequence of simpler sub-problems before producing a final answer. Wei *et al.* (2022) demonstrate substantial accuracy improvements on arithmetic, commonsense, and multi-hop question answering benchmarks, with the capability emerging at approximately 100 billion parameters. Zero-shot chain-of-thought (Kojima *et al.*, 2022) achieves similar benefits without worked examples by adding the phrase "Let's think step by step" to the prompt, demonstrating that the underlying capability is latent in sufficiently large models and requires only elicitation.

The limitation of CoT is that each intermediate step is generated by the same autoregressive mechanism described in the previous section. An incorrect intermediate step propagates forward as conditioning context, and no mechanism verifies that any step corresponds to a fact in an external knowledge source. A CoT chain can be entirely fluent and internally coherent while containing one or more steps that are factually fabricated.

2.4.2 Extensions: Self-Consistency and Tree-of-Thought

Two major extensions of chain-of-thought address its reliability limitations without resolving its structural faithfulness problem. Self-consistency (Wang *et al.*, 2023) samples multiple independent CoT chains and selects the final answer by majority vote, marginalizing over individual chain errors. This approach substantially improves accuracy on benchmarks such as GSM8K and multi-hop QA by reducing sensitivity to individual hallucinated chains. However, it provides no guarantee that any single chain is factually grounded: a majority of independently hallucinated chains can still produce an incorrect majority-vote answer, particularly on deep multi-hop questions where error probabilities compound.

Tree-of-Thought (Yao *et al.*, 2023) frames generation as a search over a tree of partial reasoning sequences, using the language model as both a generator and an evaluator to prune unpromising branches. This improves performance on tasks requiring backtracking and exploration. The fundamental limitation remains: pruning decisions are made by the model itself, using its own parametric knowledge as the sole evaluator, without any external verification of the factual status of candidate reasoning steps.

2.4.3 The Faithfulness Ceiling of Prompting-Based Approaches

The common limitation of all prompting-based approaches, including few-shot, chain-of-thought, self-consistency, tree-of-thought, and their combinations, is that they modify the *input* to the language model without modifying the *generation mechanism*. Because decoding still occurs over the full vocabulary, every token retains a strictly positive probability at every step. True structural faithfulness, defined as the guarantee that every generated token corresponds to a verified external fact, requires that invalid tokens be assigned exactly zero probability. This can only be achieved by intervening directly in the decoding algorithm, which is the function of constrained decoding approaches discussed in Section 2.6 (Geng *et al.*, 2023; Willard and Louf, 2023).

2.5 Knowledge Graphs and Knowledge Graph Question Answering

2.5.1 Knowledge Graphs

A knowledge graph is a structured database of verified facts, represented as a set of directed, labelled triplets (e, r, e') where $e, e' \in \mathcal{E}$ are entities and $r \in \mathcal{R}$ is a relation type. Each triplet encodes a single human-verified fact: for example, (*Marie Curie*, *award_received*, *Nobel_Prize_Physics*) or (*Blue_Hawaii*, *film.film.featured_film_location*, *Hawaii*). Freebase (Bollacker *et al.*, 2008), the knowledge graph underlying the evaluation benchmarks used in this work, encodes tens of millions of such triplets across geography, history, entertainment, and science, and has been widely adopted as the standard resource for KGQA research (Yih *et al.*, 2016b; Talmor and Berant, 2018b).

The critical distinction between a knowledge graph and an unstructured text corpus is that every triplet in the KG has been explicitly verified and is machine-readable in a deterministic, unambiguous form. This property makes KGs suitable as sources for hard constraints on

generation: a candidate token can be verified against the KG via an exact lookup, rather than via probabilistic semantic matching against raw text. A multi-hop reasoning path of length L through the KG is a sequence $(e_0, r_1, e_1, \ldots, r_L, e_L)$ in which each consecutive triplet (e_{i-1}, r_i, e_i) is a verified member of $\mathcal{T}$ (Lan *et al.*, 2022b).

2.5.2 Knowledge Graph Question Answering

Knowledge graph question answering (KGQA) requires a system to identify the answer entity e_L to a natural language question q by reasoning through a chain of verified KG triplets. The task is formally defined as: given question q with associated question entities $\mathcal{E}_q \subseteq \mathcal{E}$ identified by entity linking, identify the terminal entity of the correct reasoning path in $\mathcal{G}$ (Lan *et al.*, 2022b).

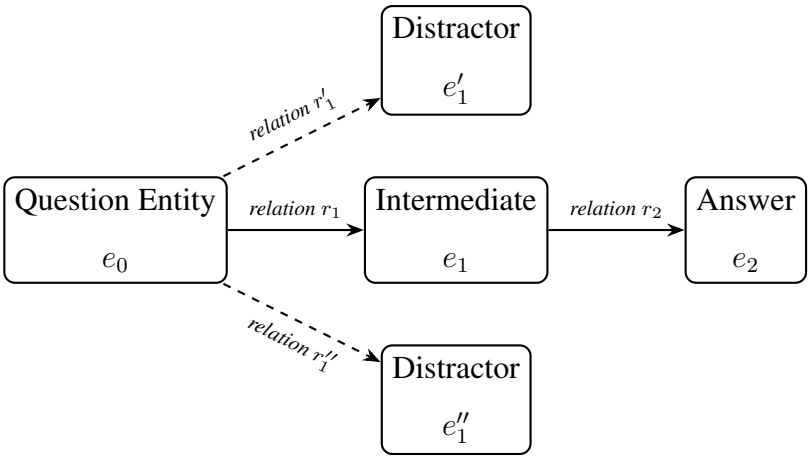

Figure 2.2 A two-hop reasoning path in a knowledge graph.

Early KGQA approaches relied on either semantic parsing (converting questions to structured SPARQL queries) or information retrieval over KG subgraphs (Lan *et al.*, 2022b). More recent approaches use large language models as retrievers, agents iteratively querying the KG, or generators conditioned on retrieved subgraph context (Jiang *et al.*, 2023a; Jin *et al.*, 2024; Luo *et al.*, 2024a). The shift toward LLM-based KGQA has improved fluency and coverage but reintroduced the hallucination risk that structured approaches avoided: LLM-generated reasoning chains are not guaranteed to correspond to verified KG paths.

2.5.3 Evaluation Benchmarks

Two benchmark datasets provide the evaluation setting for KGQA research. WebQSP (Yih *et al.*, 2016b) is derived from the WebQuestions dataset and comprises 4,737 questions grounded

in Freebase with reasoning depths of one to two hops, annotated with SPARQL queries and evaluated using Hits@1 and F1. ComplexWebQuestions (CWQ; Talmor and Berant 2018b) extends WebQSP by compositionally transforming questions through conjunction, composition, and comparative operations, yielding approximately 34,000 questions requiring up to four reasoning hops. WebQSP represents a regime where the set of valid paths is comparatively small; CWQ represents the regime where valid path sets grow large and the difficulty of identifying the correct path among structurally valid alternatives is highest.

2.5.4 Multi-Hop Complexity

The combinatorial difficulty of multi-hop KGQA increases sharply with hop depth. For an entity with average out-degree d in the KG, the number of valid paths of length L reachable from the question entities grows as $|\mathcal{E}_q| \times d^L$. For well-connected Freebase entities where $d \approx 20$ and a question with up to three question entities, a three-hop question can produce up to 24,000 structurally valid paths, of which exactly one leads to the correct answer (He *et al.*, 2021b; Lan *et al.*, 2022b). This combinatorial explosion has concrete consequences for any approach that must discriminate among candidate paths: at each generation step, the model must select the correct next token from among a very large set of valid-but-incorrect alternatives, relying on its parametric knowledge to do so.

2.6 Constrained Decoding for Faithful Text Generation

Constrained decoding is the paradigm in which generation is restricted to a predefined valid set at each step through direct intervention in the decoding mechanism. Unlike prompting, which modifies the input to the LLM, constrained decoding modifies the token selection process itself: invalid tokens are assigned zero probability before sampling, making their generation structurally impossible.

2.6.1 Logit Masking

The foundational mechanism of constrained decoding is logit masking, described formally in Geng *et al.* (2023) and Willard and Louf (2023). A constraint oracle computes a binary validity mask over the vocabulary at each step; the mask is applied element-wise to the raw logit vector before softmax normalization, setting invalid token logits to negative infinity and thereby assigning them zero probability. Token selection then operates exclusively over

the valid subset. The central design question in any constrained decoding system is how the validity mask is computed: what information the oracle uses, and how it changes across generation steps.

2.6.2 Grammar-Constrained Decoding

Geng *et al.* (2023) introduce Grammar-Constrained Decoding (GCD), which constrains generation to strings conforming to a context-free grammar, using an Earley parser as the oracle. The valid next-token set at each step is the set of tokens that extend the current partial string as a valid grammar prefix. GCD additionally introduces input-dependent grammars, in which the grammar is parameterized by the input question, representing a partial step toward context-sensitive constraints. Willard and Louf (2023) address the computational cost of constrained decoding with an efficient finite-state machine formulation. By precomputing an index from automaton states to valid vocabulary subsets, the valid token set can be retrieved in amortized constant time at each generation step, making constrained decoding practical at inference speed.

The shared limitation of these format-constraint approaches is that they enforce structural validity of the output format (such as grammatical correctness or schema conformance) without enforcing factual correctness. A grammatically valid output can contain entirely hallucinated facts. The transition from format constraints to factual constraints requires a constraint oracle that verifies tokens against a knowledge source rather than against a grammar.

2.6.3 Knowledge Graph-Constrained Decoding

The application of constrained decoding to knowledge graph faithfulness is a more recent development. The core idea is to use the KG itself as the constraint oracle: at each generation step, only tokens that correspond to valid continuations of a KG reasoning path are admitted. This approach guarantees structural faithfulness, meaning every generated token corresponds to a verified KG triplet, which makes hallucination of KG facts structurally impossible.

Graph-Constrained Reasoning (GCR) (Luo *et al.*, 2024b) is the state of the art in this line of work. GCR constructs a KG-Trie (a prefix tree over all KG paths within L hops of the question entities, built by breadth-first search before generation begins) and uses it as the

constraint oracle during beam-search decoding. The trie is queried at each generation step to return valid next tokens. GCR achieves 92.6 Hits@1 on WebQSP and 75.8 on CWQ with verified 100% structural faithfulness using a Llama-3.1-8B backbone. A key property is that the trie is constructed once before generation and never modified: the valid token set is fixed regardless of what has been generated.

Decoding on Graphs (DoG) (Li *et al.*, 2024) introduces step-wise trie expansion as a partial remedy to GCR's static oracle. DoG defines *well-formed chains* as sequences of KG triplets in which every triplet exists in the KG and every entity is connected to the preceding triplet or is a question entity. It enforces this definition through step-wise expansion: after committing to an entity, the constraint trie is extended with all KG neighbors of that entity. This makes the oracle topologically dynamic, growing as reasoning progresses through the graph. DoG achieves 90.99 Hits@1 on WebQSP and 76.08 on CWQ with 100% structural faithfulness, and is training-free, requiring no fine-tuning. However, the expansion rule admits all KG neighbors of each visited entity regardless of their relevance to the question or the current reasoning direction; and per-step trie reconstruction sacrifices the efficiency of precomputed constraint structures.

RouterKGQA (Yuan *et al.*, 2026) takes a different approach, routing between a specialised KG-reasoning model and a general LLM based on question complexity. It applies semantic filtering using SBERT embeddings (specifically nomic-embed-text-v1) to select among candidate answers after generation, achieving competitive accuracy with reduced inference cost. RouterKGQA's semantic scoring operates at the answer selection level rather than at the token constraint level: paths are generated first and filtered retrospectively, rather than being constrained during generation.

2.6.4 Beam Search and Trie-Based Constraints

To understand how knowledge graph constraints are practically enforced during generation, it is essential to examine the two primary search algorithms and data structures involved: beam search and the prefix trie.

Beam Search. First introduced in the context of the HARPY speech recognition system (Lowerre, 1976), beam search is a heuristic search algorithm used in natural language processing to decode sequences from probabilistic models. Unlike greedy decoding, which exclusively

selects the single most probable token at each step and risks generating sub-optimal global sequences, beam search maintains a set of the k most promising partial hypotheses (or “beams") at any given time. At each generation step, the algorithm expands all k current hypotheses by scoring their possible continuations, ranks the resulting sequences based on their cumulative log-probability, and prunes the search space back down to the top k candidates (Holtzman *et al.*, 2020). This approach mitigates the risk of early compounding errors while remaining computationally tractable, making it the dominant decoding strategy for tasks requiring structured and coherent output.

The Prefix Trie. A trie, originating from the concept of a “trie memory" (retrieval) introduced by Fredkin (1960), is a deterministic tree-based data structure used to store a dynamic set of sequences, typically strings. In a trie, no single node stores a complete string; instead, a node’s position in the tree defines the exact prefix it represents. The branches extending from any given node represent the deterministic set of valid next characters or tokens. In the domain of constrained decoding, a pre-computed trie serves as a highly efficient constraint oracle. Given the current generated sequence prefix, traversing the trie to the corresponding node yields the exact subset of valid next tokens in amortised constant time, $O(1)$, relative to vocabulary size. This operation is critically more efficient than running continuous dynamic parser evaluations, making trie lookups the most scalable pattern for enforcing hard factual constraints at inference speed (Willard and Louf, 2023).

The KG-Trie. For knowledge graph-constrained generation, architectures such as Graph-Constrained Reasoning (GCR) utilize a specialized adaptation known as the KG-Trie (Luo *et al.*, 2024b). A KG-Trie serializes structurally verified multi-hop paths from the knowledge graph into string formats and maps them across the language model’s precise token vocabulary. Rather than nodes representing single alphabetical characters, each edge in the KG-Trie represents a specific valid token in the LLM’s vast vocabulary space.

When beam search is merged with a KG-Trie oracle (as visualized in Figure 2.3), the decoding process transforms into a strictly bounded traversal of the prefix tree. At each autoregressive step, the current beam hypotheses map to specific states in the KG-Trie. The language model calculates a probability distribution over its entire vocabulary, but a mathematical mask forces the probability of any candidate token not validly present as a child node in the trie to exactly zero ($-\infty$ prior to softmax normalization). The beam search then advances strictly down the

permissible branches, functionally restricting the LLM's autoregressive generation to a set of pre-verified deterministic paths while still allowing the model to express internal preference (via log-probabilities) among those factually valid chains.

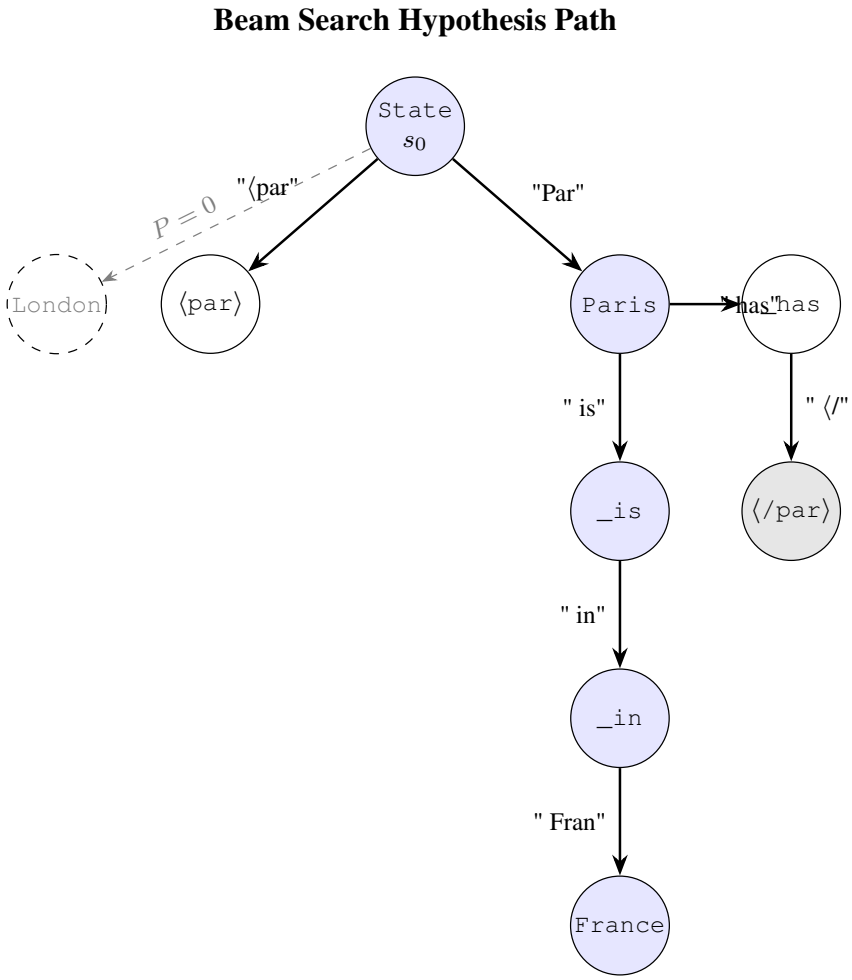

Figure 2.3 Visualization of KG-Trie constrained generation.

2.7 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG), introduced by Lewis *et al.* (2020a), is the dominant paradigm for grounding LLM outputs in external knowledge. A dense retriever selects relevant passages from an external source based on embedding similarity, and a reader LLM generates the answer conditioned on the retrieved context and the question. Fusion-in-Decoder (Izacard and Grave, 2021) extends this by encoding multiple retrieved passages independently and fusing them in the decoder, improving coverage.

2.7.1 KG-Augmented Retrieval

Several works augment RAG with KG-structured context. He *et al.* (2021b) retrieve KG subgraphs and use attention over retrieved facts to guide multi-hop reasoning. Yasunaga *et al.* (2021) combine retrieved text passages and KG neighborhoods via graph neural networks. Shi *et al.* (2021) retrieve KG paths as natural language context strings, providing path-shaped input to the reader. G-Retriever (He *et al.*, 2024) applies RAG specifically to textual graph understanding, retrieving graph-structured context for QA tasks.

2.7.2 Advanced RAG Variants

More recent RAG variants address the timing and precision of retrieval. Self-RAG (Asai *et al.*, 2024) inserts reflection tokens that allow the model to dynamically decide when retrieval is needed and whether retrieved passages support the claim being generated, improving retrieval precision and citation accuracy. FLARE (Jiang *et al.*, 2023b) uses the model's next-token prediction confidence as a retrieval trigger, retrieving fresh context when confidence falls below a threshold. IRCoT (Trivedi *et al.*, 2023) interleaves retrieval with chain-of-thought steps, retrieving relevant context at each intermediate reasoning step and improving multi-hop accuracy substantially. GraphRAG (Edge *et al.*, 2024) constructs community-structured knowledge graphs from document corpora and retrieves community summaries for complex relational queries, explicitly leveraging graph structure for improved coverage.

2.7.3 The Structural Limitation of Soft Grounding

Despite their empirical improvements, all RAG approaches share a structural limitation: retrieved context is a soft influence on generation. The retriever determines what the LLM sees; it does not determine what the LLM is permitted to generate. Because the LLM's generation mechanism remains unchanged (the distribution over the full vocabulary is computed autoregressively), there exists a nonzero probability of generating any token at any step, including tokens that correspond to no verified KG fact. Structural faithfulness with probability one, which guarantees that every generated token corresponds to a verified fact, cannot be achieved under any architecture that modifies only the input context without modifying the token selection mechanism (Luo *et al.*, 2024b; Geng *et al.*, 2023).

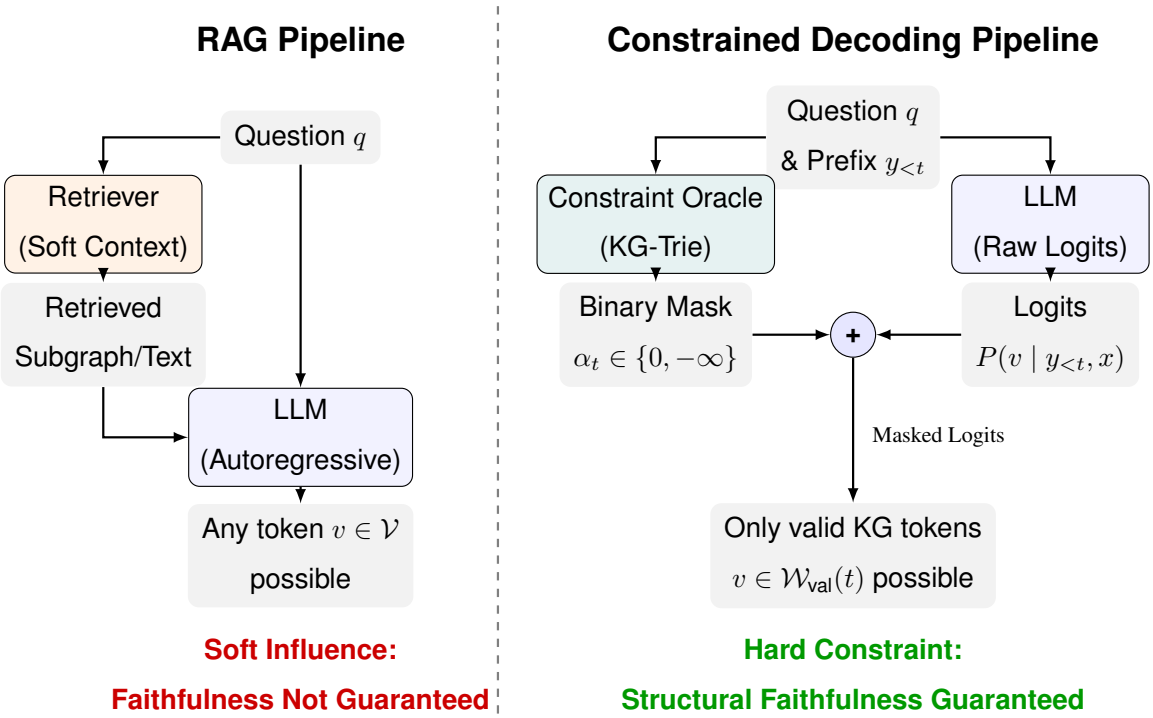

Figure 2.4 Architectural comparison of standard Retrieval-Augmented Generation (left) and KG-constrained decoding (right).

2.8 Semantic Relevance Scoring with Sentence Transformers

2.8.1 Sentence Transformer Embeddings

Sentence-BERT (Reimers and Gurevych, 2019b) introduced a class of sentence encoders trained to produce dense vector representations in which semantically similar sentences cluster together and semantically dissimilar sentences are pushed apart. The SBERT architecture applies mean pooling over token representations from a BERT-like transformer encoder, producing a fixed-length embedding. Training uses a siamese network on labelled sentence pairs with a classification objective over entailment, neutral, and contradiction labels. The resulting embedding space supports cosine similarity as a reliable proxy for semantic alignment between sentences, without requiring task-specific fine-tuning at inference time.

2.8.2 Applications in Retrieval and Graph Reasoning

Sentence transformer similarity has been applied extensively in dense retrieval settings. Karpukhin *et al.* (2020) demonstrate that a bi-encoder architecture trained on question-passage pairs produces embeddings that outperform sparse retrieval (BM25) on open-domain QA. Xiong *et al.* (2021) refine bi-encoder training, improving coverage of relevant passages.

In the KGQA context, Shi *et al.* (2021) and Saxena *et al.* (2022) apply semantic similarity to rank candidate KG paths for retrieval, selecting high-scoring paths as soft context for the reader LLM. These approaches use cosine similarity as a soft ranking signal over candidate paths, selecting the top-k for context provision.

2.8.3 Encoder Architecture Trade-offs

The choice of sentence encoder involves a quality-latency tradeoff that is particularly stringent in step-wise constrained decoding, where the encoder is invoked at each generation step. Larger encoders such as all-mpnet-base-v2 (110M parameters) and BGE-M3 (Chen *et al.*, 2024) (568M parameters) achieve higher performance on semantic textual similarity benchmarks but incur substantially higher inference latency. Lexical approaches such as BM25 are fast but cannot capture semantic similarity between paraphrase-equivalent paths that share no surface tokens. The all-MiniLM-L6-v2 model (Reimers and Gurevych, 2019b), at 22M parameters with approximately 80ms per-query CPU latency, provides the optimal balance of semantic quality and inference speed for iterative step-wise applications.

2.9 Related Works

The following sections organise prior work into thematic clusters, describing what each cluster of approaches achieves, the shared mechanisms across works within each cluster, and the limitations that motivate further research.

2.9.1 Knowledge Graph-Enhanced LLM Reasoning

A large body of work explores how LLMs can be enhanced with KG access for reasoning and question answering without enforcing structural faithfulness constraints. Jiang *et al.* (2023a) introduce StructGPT, a general framework enabling LLMs to reason over structured data including KGs through an iterative reading-and-reasoning procedure. Jin *et al.* (2024) propose Graph Chain-of-Thought, augmenting LLM CoT reasoning with graph-structured evidence retrieved at each reasoning step. Luo *et al.* (2024a) introduce Reasoning on Graphs (RoG), which generates reasoning plans over KG paths and executes them faithfully, achieving strong multi-hop accuracy while maintaining path interpretability. Jiang *et al.* (2022) propose UniKGQA, unifying retrieval and reasoning for multi-hop KGQA through a shared parameter space.

These works collectively demonstrate the value of explicit KG access for improving LLM reasoning quality and interpretability. Their shared limitation is that KG access is advisory rather than constraining: the LLM attends to retrieved KG context but is free to generate beyond it. The reasoning chains produced may be plausible and KG-informed without being guaranteed to correspond to verified KG paths. Luo *et al.* (2024b) measure this gap directly, finding that RoG-style approaches produce hallucinated reasoning steps in approximately 33% of cases despite explicit KG access.

2.9.2 Retrieval-Based KGQA

Earlier KGQA approaches relied on structured retrieval rather than LLM generation. He *et al.* (2021b) propose a method for multi-hop KGQA that learns intermediate supervision signals to guide path selection through the KG, showing that explicit intermediate supervision improves performance at greater hop depths. Yasunaga *et al.* (2021) combine text and KG context in a joint graph neural network, propagating information between retrieved passage tokens and KG entity nodes. Shi *et al.* (2021) retrieve KG paths by semantic similarity and provide them as natural language context to a reader model. Saxena *et al.* (2022) apply entity and relation embeddings to select multi-hop paths for KGQA without LLM generation.

These retrieval-based approaches avoid LLM hallucination by operating within the structured KG rather than generating freely. Their limitation is expressiveness: they are typically constrained to fixed path structures and do not leverage the multi-step reasoning capability of modern LLMs. The transition to LLM-based approaches improved naturalness and coverage at the cost of faithfulness guarantees.

2.9.3 Format-Constrained and Entity-Constrained Generation

De Cao *et al.* (2021) introduce autoregressive entity retrieval, constraining generation to a finite set of entity names using a trie over known entity strings. This approach ensures that generated entities correspond to real entities in a fixed vocabulary but does not enforce path-level faithfulness across multiple steps. Geng *et al.* (2023) generalise this to arbitrary context-free grammar constraints, while Willard and Louf (2023) provide an efficient finite-state automaton formulation. Deutsch *et al.* (2019) propose a general algorithm for constrained sequential inference applicable across structured prediction tasks.

These works establish that hard constraints on generation are computationally tractable and empirically beneficial. Their limitation for KGQA is that they enforce output *format* rather than *factual content*: constraining generation to grammatically valid strings does not ensure that the strings correspond to verified KG paths. The extension from format constraints to KG path constraints requires a constraint oracle that encodes the relational structure of the knowledge graph.

2.9.4 Semantic Similarity in Graph Reasoning

Several works apply semantic similarity signals to guide reasoning over knowledge graphs. Shi *et al.* (2021) rank candidate KG paths by semantic similarity between the question and path descriptions, retrieving high-scoring paths as evidence. Saxena *et al.* (2022) learn embeddings for KG entities and relations that support similarity-based multi-hop path selection. He *et al.* (2021b) use attention-guided traversal over retrieved KG subgraphs, with attention weights reflecting the relevance of each subgraph region to the question. PathRetriever (Asai *et al.*, 2020) learns to retrieve reasoning paths from evidence graphs by scoring paths using a recurrent model over their node sequences.

These works demonstrate that semantic similarity can effectively differentiate relevant from irrelevant KG paths in a retrieval setting. The key architectural distinction is that semantic similarity is applied as a *soft ranking signal*: candidate paths are scored and the highest-scoring ones are provided to the LLM as context or used to select among generated candidates. No existing work applies semantic similarity as a *hard binary admission criterion* that determines which tokens the LLM is permitted to generate at each decoding step. The distinction between soft retrieval signals and hard generation constraints is significant: the former influences what the LLM sees, while the latter determines what the LLM can produce.

2.9.5 Hallucination Mitigation in Knowledge-Intensive Tasks

Beyond the constrained decoding paradigm, a range of approaches attempt to reduce hallucination in knowledge-intensive tasks. Retrieval-augmented approaches (including RAG (Lewis *et al.*, 2020a), Self-RAG (Asai *et al.*, 2024), FLARE (Jiang *et al.*, 2023b), IRCoT (Trivedi *et al.*, 2023), and GraphRAG (Edge *et al.*, 2024)) address the problem by providing relevant external context, improving factual coverage without modifying the generation mechanism. Wang *et al.* (2023) improve consistency by majority-voting over multiple CoT chains. Huang

et al. (2024) study whether LLMs can self-correct reasoning errors, finding that self-correction is generally ineffective without external feedback, reinforcing that internal parametric knowledge cannot reliably detect its own errors.

Post-hoc verification approaches (Nakano *et al.*, 2021; Gao *et al.*, 2023) check generated claims against external sources after generation and revise or flag unsupported statements. While effective at identifying errors, these approaches require a separate verification pass and do not prevent hallucination during generation. Agrawal *et al.* (2024a) systematically analyse failure modes in RAG systems, finding that even with relevant context provided, LLMs regularly produce statements unsupported by the retrieved evidence.

2.9.6 Concurrent Work

RouterKGQA (Yuan *et al.*, 2026) is the work most closely related to this research in terms of shared components. RouterKGQA routes KGQA queries between a specialised KG reasoning model and a general LLM based on question complexity, and applies SBERT-based semantic similarity (using the nomic-embed-text-v1 encoder) to filter candidate reasoning paths. It achieves competitive performance with reduced inference cost, averaging approximately 1.15 LLM calls per question.

The architectural distinction from the approach in this dissertation is the level at which semantic scoring is applied. RouterKGQA scores complete reasoning paths after they have been generated, using similarity to filter among candidates retrospectively. This approach does not modify the token selection mechanism during generation: the LLM generates freely, and semantic similarity is applied post-hoc to select among the outputs. In the constrained decoding paradigm, by contrast, semantic similarity would need to operate as a hard admission criterion applied before or during token selection, determining which tokens the LLM is permitted to generate at each step. RouterKGQA does not provide a metric for constraint permissiveness and does not measure the degree to which its candidate sets contain structurally valid but semantically irrelevant paths.

2.10 Synthesis: Three Unresolved Gaps

The preceding review reveals that two distinct research trajectories (constrained decoding for structural faithfulness, and semantic similarity for relevance-guided reasoning) have developed largely in parallel without being integrated at the constraint-oracle level. Constrained decoding approaches (Luo *et al.*, 2024b; Li *et al.*, 2024) achieve structural faithfulness but construct their constraint oracles without reference to question semantics or generation context, producing valid token sets that contain large numbers of irrelevant paths. Semantic similarity approaches (Shi *et al.*, 2021; Saxena *et al.*, 2022; Yuan *et al.*, 2026) improve relevance but operate as soft signals that influence retrieval or post-hoc selection without modifying the generation mechanism.

Three specific gaps emerge from this review.

Gap 1 – The Permissiveness Problem. Existing KG-constrained decoding frameworks construct their constraint oracles as functions of graph structure and question entities alone, without reference to question semantics or the evolving generation state. For multi-hop questions, this produces valid token sets containing thousands of structurally correct but semantically irrelevant paths at every generation step. The LLM must discriminate among these paths using its own parametric knowledge, reintroducing the hallucination risk that constrained decoding was designed to eliminate (Luo *et al.*, 2024b; Li *et al.*, 2024).

Gap 2 – The Static-Dynamic Efficiency Tradeoff. GCR's precomputed static trie is computationally efficient but semantically blind and identical at every generation step. DoG's step-wise topological expansion introduces dynamism at the cost of per-step trie reconstruction and without semantic filtering (Li *et al.*, 2024; Willard and Louf, 2023). No existing framework achieves semantic responsiveness (where the valid set narrows progressively as reasoning commits to a specific direction) while preserving the efficiency of precomputed constraint structures.

Gap 3 – The Absence of a Constraint Quality Metric. No existing work provides a metric that quantifies constraint permissiveness independently of downstream answer accuracy. Without such a metric, it is impossible to measure progress on reducing irrelevant paths in the valid set, compare frameworks on constraint quality independently of model quality, or

diagnose permissiveness as a failure mode in deployed systems (Luo *et al.*, 2024b; Li *et al.*, 2024).

Table 2.1 Summary of key related works and gap coverage.

Work	**Approach**	**Structural faithfulness**	**Semantic conditioning**
Luo *et al.* (2024b)	Static KG-Trie decoding	Yes (100%)	No
Li *et al.* (2024)	Step-wise topological expansion	Yes (100%)	No
Yuan *et al.* (2026)	Routing + post-hoc filtering	Probabilistic	Yes (answer-level)
Luo *et al.* (2024a)	Reasoning path planning	No	Partial
Jiang *et al.* (2023a)	LLM over structured data	No	Partial
He *et al.* (2021b)	KG subgraph attention	No	Partial
Asai *et al.* (2024)	Retrieval with reflection	No	Soft
Edge *et al.* (2024)	Community-structured retrieval	No	Soft
Shi *et al.* (2021)	Path retrieval by similarity	No	Soft (retrieval)
Geng *et al.* (2023)	Grammar-constrained decoding	Format only	No
Willard and Louf (2023)	FSM-indexed constrained decoding	Format only	No

These three gaps collectively motivate an approach that integrates semantic relevance scoring directly into the constraint oracle, conditioning the valid token set jointly on the knowledge graph, the question semantics, and the evolving generation state, which is the research direction developed in the following chapters.

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